HW1 - Classification trees, random forests

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1 CART and Random Forests

Each prediction is a Bernoulli trial (correct or not). For misclassification rate $\hat{p}$ on n samples:

$$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

1.1 Classification Tree

Full tree with `min_samples=2`:

Dataset	Misclassification rate	SE
Train	0.000	0.000
Test	0.267	0.057

1.2 Random Forest

Random forest with $n = 100$ trees, `min_samples=2`:

Dataset	Misclassification rate	SE
Train	0.000	0.000
Test	0.233	0.055

1.3 Misclassification Rate vs Number of Trees

Graph shown in Fig. 1.

2 Variable Importance

Permutation-based importance: for each tree, the OOB (out-of-bag) accuracy is computed, then each feature is permuted in the OOB samples and accuracy is recomputed. The drop in accuracy averaged over all trees gives the feature's importance. The graph is shown in Fig. 2.

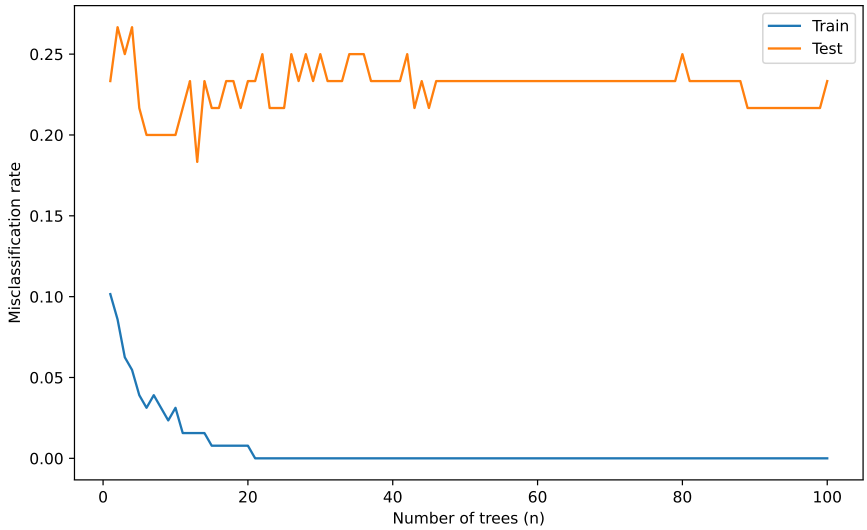

Figure 1: Misclassification Rate vs Number of Trees

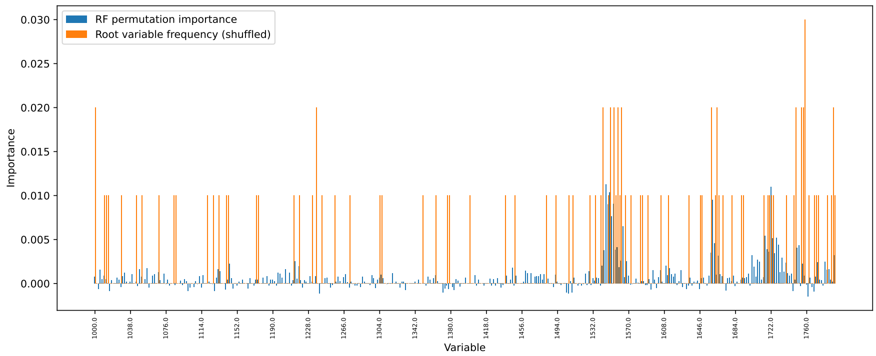

Figure 2: Variable importance

3 Improved Trees

All results below use 10-fold cross-validation on the combined train+test data (188 samples).

Method	CV error	SE
Tree (`min_samples=2`)	0.208	0.022
BetterTree (`min_samples=5`)	0.197	0.024
BetterTree2 (PCA, 20 components)	0.227	0.029

3.1 BetterTree: Pre-pruning

The full tree with `min_samples=2` achieves 0% training error but 20.8% CV error, indicating overfitting. With 396 features and only ∼170 training samples per fold, the tree can always find a feature that perfectly separates a small node—even when that split captures noise rather than signal. Increasing `min_samples` to 5 prevents the tree from creating leaves with very few samples, forcing it to generalize. This yields a modest but consistent improvement (19.7% vs 20.8%).

We also tested cost-complexity pruning (CART's standard post-pruning) and reduced error pruning, but both performed worse. Cost-complexity pruning requires nested CV for α selection, which is unreliable with only ∼170 samples. Reduced error pruning wastes 20% of already scarce data on a validation split. On small datasets, the simplest regularization (larger `min_samples`) outperforms more sophisticated but data-hungry pruning strategies.

3.2 BetterTree2: PCA + tree (negative result)

FTIR spectra have highly correlated adjacent features, making PCA a natural preprocessing step. However, PCA projects onto directions of maximum variance, which may not align with the directions most discriminative for classification. Trees split on individual features, and PCA components are dense linear combinations of all features—this mismatch means PCA can obscure the sharp spectral peaks that a tree would otherwise exploit. Performance degraded across all tested component counts (3–50), confirming that for tree-based classifiers on spectral data, retaining the original feature space is preferable.

3.3 Trade-offs

The fundamental limitation is that a single decision tree cannot escape the bias-variance trade-off: with few samples and many features, either the tree overfits (low `min_samples`) or underfits (high `min_samples`). The random forest's advantage—averaging over diverse trees—is the principled solution, which is why it consistently outperforms any single-tree approach on this data.